

Statistical Mechanics (MP461) Assignment 1

Due Friday 2nd of October 2026

0.0.1 1. Quanta of energy

A system contains n atoms, each of which can only have zero or one quanta of energy. How many ways can you arrange r quanta of energy when (a) $n = 2, r = 1$; (b) $n = 20, r = 10$ and (c) $n = 2 \times 10^{23}, r = 1 \times 10^{23}$?

0.0.2 2. Fractional error

What fractional error do you make when using Stirling's approximation (in the form $\ln n! \approx n \ln n - n$) to evaluate (a) $\ln 10!$, (b) $\ln 100!$, and (c) $\ln 1000!$?

3. A fair die

A throw of a regular die yields the numbers 1, 2, ..., 6, each with probability 1/6.

1. Find the mean, variance, and standard deviation of the numbers obtained.
2. Two throws of a die yield a total of 2 up to 12. What is the mean, variance, and standard deviation in this case?
3. Calculate the probability of a total of 13 after 4 throws.
4. What about the probability of a 14 after 4 throws?

Hint: For the last 2 you could first consider the 2-throw probabilities

4. Poisson Distribution

This question is about a discrete probability distribution known as the Poisson distribution. Let x be a discrete random variable that can take the values 0, 1, 2, ... A quantity x is said to be Poisson distributed if the probability $P(x)$ of obtaining x is

$$P(x) = \frac{e^{-m} m^x}{x!} \quad (1)$$

where m is a particular number (which you will show in part (b) of this exercise is the mean value of x)

(a) Show that $P(x)$ is a well-behaved probability distribution in the sense that

$$\sum_{x=0}^{\infty} P(x) = 1 \quad (2)$$

(Why is this condition important?)

(b) Show that the mean value of the probability distribution is

$$\langle x \rangle = \sum_{x=0}^{\infty} x P(x) = m. \quad (3)$$

5. Exponential distribution

Let x be a continuous random variable that can take any value $x \geq 0$. A quantity is said to be exponentially distributed if it takes on values between x and $x+dx$ with probability:

$$P(x)dx = Ae^{-x/\lambda}dx, \quad (4)$$

where λ and A are constants.

- (a) Find the value of A that makes $P(x)$ a well defined probability distribution.
- (b) Show that the mean value of the probability distribution is λ .
- (c) Find this probability distribution's variance and standard deviation.

6. Moment Generating Functions

This question discusses an efficient method for calculating the mean and variance of probability distributions. We define the **moment generating function** $M(t)$ for a random variable x by

$$M(t) = \langle e^{tx} \rangle. \quad (5)$$

- (a) Show that this definition implies that

$$\langle x^n \rangle = M^{(n)}(0), \quad (6)$$

where $M^{(n)}(t) = d^n M/dt^n$, the mean $\langle x \rangle = M^{(1)}(0)$ and the variance $\sigma_x^2 = M^{(2)}(0) - [M^{(1)}(0)]^2$.

- (b) Show that for the Poisson distribution, $M(t) = e^{m(e^t-1)}$ agrees with the mean calculated earlier.
- (c) Use this method to work out the variance of the Poisson distribution.

7. Random Walk in One Dimension

A particle starts at position $x = 0$ and takes a random walk along the x -axis. At each step, the particle moves either $+1$ or -1 with equal probability ($1/2$ each).

- (a) After 4 steps, what is the probability that the particle is at position $x = 0$?
- (b) After 4 steps, what is the probability that the particle is at position $x = +2$?
- (c) In general, after N steps, what is the probability $P(x, N)$ that the particle is at position x ? (Assume N and x have the same parity, i.e., both even or both odd)
- (d) Investigate the behavior of $P(x, N)$ for large N . What type of distribution does it approach, and what can you conclude about the typical distance travelled after many steps?

Hint: Use Stirling's approximation for large factorials. Also, for the expansion, you may find it useful that $\ln(1 + \epsilon) \approx \epsilon - \frac{\epsilon^2}{2}$ for small ϵ .